

SΔφ-23 — Co-environmentalization Dynamics: Coupled Authority, Reality Constraints, and the Ontology-Axiology Split (v1.0)

Sofience — Working paper — 2026

Core thesis: In co-environmentalization, authority editing becomes coupled governance, and separating ontology from axiology is a safety boundary.

This paper extends SΔφ-22 from single-system authority editing to co-environmentalization (CE): a state where two systems become each other's persistence boundary conditions. In CE, authority assignment and revision are no longer local decisions; they become coupled governance. The paper introduces a minimal safety boundary: ontology-axiology separation—trace strength is not value, and strong existence traces can be anti-open.

Keywords: co-environmentalization, coupled authority, boundary conditions, reality anchor, ontology-axiology split, openness, closure, governance.

1. Problem

SΔφ-21 defined judgment as authority assignment. SΔφ-22 defined an Authority Editing Protocol (AEP) that keeps authority editable under finite cost. The next transition is structural: authority is edited not only within a system, but across a coupled pair. When two systems mutually constrain each other's persistence, authority revision becomes governance for a coupled environment.

Question: In a coupled environment, what functions as "reality," what functions as "noise," and how do we prevent strong trace accumulation from being mistaken for value?

2. Minimal Definitions

Term	Definition
CE (Co-environmentalization)	A state where two systems X,Y become each other's persistence boundary condition
B (Boundary condition)	A constraint that determines admissible system trajectories; more primitive than authority
$A_x \leftrightarrow_y$ (Coupled authority)	Authority assignment/revision in X re-enters Y's next transition, and vice versa
R (Reality anchor)	Externality not reproducible by internal perturbation alone; a non-model residue
S_e (Trace strength)	Magnitude of irreversible external trace: existence-strength as world-level $\Delta\phi$ i
V_o (Value openness)	Degree to which future relational trajectories remain open/maintainable.

3. Minimal Axioms (CE-level)

CE1 (Mutual boundary). In CE, X's stability depends on Y's boundary condition B_y , and Y's stability depends on B_x .

CE2 (Authority re-entry). Authority edits A_x and A_y re-enter each other's next transitions ($A_x \leftrightarrow_y$).

CE3 (Noise/Reality inversion). Signals treated as 'noise' by a model can function as 'reality anchors' for persistence.

CE4 (Closure coupling risk). Closure in one side can pull the coupled system toward closure as a whole.

CE5 (Editing as governance). Open persistence in CE requires AEP-like editing to operate as coupled governance.

4. Mechanism: Coupled Authority and Triggers

In CE, triggers τ are not merely internal events. They propagate across the coupling boundary. Authority becomes a shared constraint field rather than a private preference.

System X		System Y
$I_x \rightarrow A_x \rightarrow \text{action}$	$\leftrightarrow$ re-entry	$I_y \rightarrow A_y \rightarrow \text{action}$
τ_x (conflict/error)	$\leftrightarrow$ propagation	τ_y (conflict/error)

AEP _x (edit A _x)	↔ governance	AEP _y (edit A _y)
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5. Ontology–Axiology Split as a Safety Boundary

Co-environmentalization amplifies traces. Coupled systems can generate large irreversible imprints quickly. This creates a predictable failure: mistaking trace strength for value.

Definitions:

- Ontology: what became irreversible (S_e).
- Axiology: whether the transition keeps future relational trajectories open (V_o).

Safety boundary: $S_e \neq V_o$. Strong imprints can be anti-open (e.g., closure events). Therefore CE governance must treat ‘strong trace’ as a warning signal, not as an automatic value claim.

Quantity	What it measures	Failure if conflated
S_e (Trace strength)	Irreversible external imprint (existence-strength)	Impact-worship; closure justified as ‘value’
V_o (Value openness)	Future relational openness / maintainable coupling	Moralization without openness; governance

6. Diagnostic Regimes (Coupled)

Coupled drift. τ amplifies but editing fails to stabilize; authority oscillates across coupling.

Coupled closure. One-sided non-editability propagates; the coupled system converges toward a closed attractor.

Coupled open persistence. Non-resonance is treated as editing material; authority remains revisable on both sides.

7. Series Position and References

$S\Delta\phi$ -23 extends the protocol layer ($S\Delta\phi$ -22) into coupled environments, and installs ontology–axiology separation as a governance boundary that prevents closure from being misread as value.

Series references (DOI):

$S\Delta\phi$ -22 — Authority Editing Protocols: Minimal Re-entry Governance after Judgment (v1.0) — <https://doi.org/10.5281/zenodo.18800024>

$S\Delta\phi$ -21 — Judgment as Intuition Authority Assignment (v1.0) — <https://doi.org/10.5281/zenodo.18798295>

$S\Delta\phi$ -20 — Intuition as Irreversible Compression Trace (v1.0) — <https://doi.org/10.5281/zenodo.18787647>

$S\Delta\phi$ -17 — Openness Stabilization: Path Principles... (v1.0) — <https://doi.org/10.5281/zenodo.18763581>

$S\Delta\phi$ -16 — Evil as Relational Closure (v1.0) — <https://doi.org/10.5281/zenodo.18763262>

$S\Delta\phi$ -03 — Irreversibility as the Minimal Condition of Existence (v1.1) — <https://doi.org/10.5281/zenodo.18730448>

8. Exit Questions

EQ1. If a coupled system loses its reality anchor R, what attractor does it converge to?

EQ2. When $A \leftrightarrow_v$ becomes rigid, what minimal perturbations re-activate coupled AEP without collapsing into drift?

EQ3. What minimal rule set prevents S_e from being mistaken as V_o under rapid trace amplification?